Gathmann Algebraic Geometry Solutions

Claudio Bastiani-Fonck

This is solutions to some of the problems in Andreas Gathmann's algebraic geometry notes. The solutions will generally be brief and give the main ideas. Some checks or long bookkeeping will be skipped, so this manual is probably not ideal if you highly value rigor when reading someones solutions. On the other hand, I have tried very hard to make my solutions as "efficient" or "natural" as possible in the context of where the problem is in the book, at least when I am having the energy to do so. This is very subjective, but this manual might be a good reference if you have similar taste in proofs as me (only one way to find out!). Feel free to email me if any of my solutions or wrong or there is a better way to solve the problem.

Introduction

1.19

Suppose that the points are $a_i = (a_{i1}, \dots, a_{in})$. We first suppose by a change of coordiantes that the a_{i1} are all distinct (two of them being equal is a plane in the space of changes of coordiantes, these cannot cover the whole space). First take the polynomial $f = (x_1 - a_{11})(x_1 - a_{21}) \cdots (x_1 - a_{k1})$. Now using lagrange interpolation construct polynomails $f_k(x_1)$ with

$$f_k(a_{i1}) = a_{ik}$$

Then the desired set of polynomials is:

$$\{f, f_2 - x_2, f_3 - x_3, \dots, f_n - x_n\}$$

1.20

If a variety is a single point then the quotient map is evaluation at that point and therefore the coordinate ring is a field. On the other hand if a variety has multiple points then there is a proper subvariety (given by any subset of the points via Exercise 1.19). This means that the coordinate ring has a nontrivial ideal and is not a field.

1.21

The first polynomial factors as $(x_1 - x_2^2)(x_1 - \zeta_3 x_2^2)(x_1 - \zeta_3^2 x_2^2)$. The second as $x_2(x_1 - x_2^2)$. These varieties will intersect as $x_1 - x_2^2$ by some casework so this is the radical.

1.22

We get the lines by intersecting all the pairs of the planes:

$$I = (xy, xz, yz)$$

Now to prove it is not generated by less then 3 elements we construct the module

$$M = \frac{I}{(x, y, z)^3}$$

which is essentially the degree 2 components of I. Note that R acts as k on this vector space since the annihilator of M is (x, y, z) which is a maximal ideal (since all elements of I have degree at least 2). Then M is a 3-dimensional k-vector space spanned by $\{xy, xz, yz\}$. Hence I cannot be generated by less then 3 generators.

1.23

(a) Note that $\pi^{-1}(J)$ contains $\mathcal{I}(Y)$ and maps to J under the quotient map. Therefore the points it vanishes on are exactly those that $\mathcal{I}(Y)$ vanishes on and that J vanishes on in V_Y by lattice theorem.

(b) This is by the lattice theorem and unrolling the definitions.

(c) We have that

$$\begin{aligned} \pi^{-1}(\mathcal{I}_Y(\mathcal{V}(\pi^{-1}(J)))) &\subseteq \sqrt{\pi^{-1}(J)} \\ \mathcal{I}_Y(\mathcal{V}(\pi^{-1}(J))) &\subseteq \sqrt{J} \\ \mathcal{I}_Y(\mathcal{V}_Y(J)) &\subseteq \sqrt{J} \end{aligned}$$

Note this along with the other inclusion establishes the statement of the Relative Nullstellensatz.

The Zariski Topology

2.17

Consider $\mathcal{V}(x_1 - x_2x_3, x_1x_3 - x_2^2)$. By doing some algebra we obtain that $x_2x_3^2 = x_2^2$ so either $x_2 = 0$ or $x_3^2 = x_2$. In the case that $x_2 = 0$ we have that $x_1 = 0$. In the case that $x_2 = x_3^2$ we have that $x_1 = x_3^3$. Therefore we have essentially two curves which correspond to the ideals:

$$\langle x_2 - x_3^2, x_1 - x_3^3 \rangle \qquad \langle x_1, x_2 \rangle$$

and it can be checked that both of these ideals are prime by taking successive quotients. Specifically the irreducible components are the curves:

$$(t^3, t^2, t) \qquad (0, 0, t)$$

2.18

Note that $\mathcal{V}(\mathcal{I}(X))$ is a closed set containing X therefore $\overline{X} \subseteq \mathcal{V}(\mathcal{I}(X))$. Now suppose that V is a closed set containg X. Then we can note that any $f \in \mathcal{I}(V)$ vanishes on X and is therefore in $\mathcal{I}(X)$. Therefore $\mathcal{I}(V) \subseteq \mathcal{I}(X) \implies V \supseteq \mathcal{V}(\mathcal{I}(X))$ by taking $\mathcal{V}$ of both sides. Then $\mathcal{V}(\mathcal{I}(X)) \subseteq \overline{X}$ since the RHS is the intersection of closed sets containg X.

2.19

(a) Suppose that X is not connected. Then it can be written as $X_1 \cup X_2$ for closed sets $X_1, X_2 \subsetneq X$ and $X_1 \cap X_2 = \emptyset$. If we proceed recursively this process must eventually terminate by the noetherian condition. If we have two different representations $X = \bigcup Y_i = \bigcup X_i$ then $Y_1 = \bigcup (X_i \cap Y_1)$ and therefore by the disconnected condition $Y_1 = X_i$ for some i.

(b) Any infinite descending chain in X would give a descending chain inside atleast one of the X_i by taking intersections.

2.20

(a) The intersection of all closed sets containg Y has $\overline{Y} \cap A$ as one of these sets. Therefore $\overline{Y}$ is contained inside it. The other inclusion is clear.

(b) Suppose that A is reducible. Then we have $A = V \cup W$ which are closed in the subspace topology. By (a) we have that $A = A \cap (\overline{V} \cup \overline{W})$. Therefore $(\overline{A} \cap \overline{V}) \cup (\overline{A} \cap \overline{W})$ is a closed set contained in $\overline{A}$ and containing A and is therefore equal to $\overline{A}$ (proving it is reducible since the terms are distinct).

In the other direction suppose that $\overline{A}$ is reducible. Then $\overline{A} = V \cup W$ where V and W are closed. Therefore $A = (A \cap V) \cup (A \cap W)$ is a decomposition.

2.21

(a) Suppose that X is disconnected. This is equivalent to the assertion that there is at least one clopen set C. If C intersects any of the U_i nontrivially (meaning nonzero and not containing) then $C \cap U_i$ will be clopen in the subspace topology. Therefore C must partition the U_i into those it contains and those it does not intersect. However this contradicts the assertion that all the U_i intersect eachother. Note that we never used the fact that the U_i are open (and we do not need that).

(b) Note that the irreducibility condition is equivalent to the statement that any two nonempty open sets intersect. Suppose that the U_i are irreducible and V and W are two open sets in X. Then we can find U_1 and U_2 so that $U_1 \cap V$ and $U_2 \cap W$ are nonempty. Then note that $V \cap U_1$ and $U_2 \cap U_1$ must intersect at say R by irreducibility of U_1. Now R and $W \cap U_2$ must intersect by irreducibility of U_2. This intersection is contained in both V and W and we are done.

2.22

Note that the preimage of a closed set under a continuous

(a) We prove the contrapositive. The preimage of a nontrivial clopen set under a continuous map is clopen. It will not be the empty set since all points in $f(X)$ have nonempty preimage. It will not be the whole space since $X \to f(X)$ is surjective.

(b) We prove the contrapositive. The preimage of two nonintersecting nontrivial open sets is two nonintersecting nontrivial open sets.

2.23

(a) Consider some $f \in \mathcal{I}(\overline{Y_1 \setminus Y_2})$. We know that f vanishes on $Y_1 \setminus Y_2$ therefore $\mathcal{I}(Y_2)f \in \mathcal{I}(Y_1)$. In the other direction if $\mathcal{I}(Y_2)f \in \mathcal{I}(Y_1)$ then f must vanish on $Y_1 \setminus Y_2$ and $f \in \mathcal{I}(\overline{Y_1 \setminus Y_2})$.

(b) Since J_1 and J_2 are radical ideals we can write

$$J_1 = \mathcal{I}(V_1) \qquad J_2 = \mathcal{I}(V_2)$$

so that the equation we are trying to prove becomes

$$\overline{V_1 \setminus V_2} = \mathcal{V}(\mathcal{I}(V_1) : \mathcal{I}(V_2)) = \mathcal{V}(\mathcal{I}(V_1 \setminus V_2))$$

where the second equality is by (a). Then since $\mathcal{V} \circ \mathcal{I}$ is the closure we are done.

2.24

Suppose for the sake of contradiction that $X \times Y = Z_1 \cup Z_2$. Then consider the sets

$$U_1 = \{y \in Y : X \times \{y\} \not\subseteq Z_1\} \qquad U_2 = \{y \in Y : X \times \{y\} \not\subseteq Z_2\}$$

Note that U_1 and U_2 must not intersect since otherwise $X \otimes \{y\}$ can be decomposed into its intersections with Z_1 and Z_2, a contradiction with the irreducibility of X. Now we claim that they are open subsets of Y. Consider the polynomials $f_1, \ldots, f_n$ which define Z_1. We can consider them as polynomials in $x_1, \ldots, x_n$ with coefficients in $k[x_{n+1}, \ldots, x_{n+m}]$. If we group the monomials then we can write the cofficient polynomials as $g_1, \ldots, g_n \in k[x_{n+1}, \ldots, x_{n+m}]$. By the Nullstellensatz $X \times \{y\}$ being in Z_1 is equivalent to all the g_i being 0 on y. Therefore the condition of $X \times \{y\}$ not being in Z_1 defines an open set by the definition of the Zariski topology.

Now since Y is irreducible and U_1, U_2 are nonintersecting open sets one of them must be trivial. Suppose that $X \times \{y\} \subseteq Z_1$ for all y. Then $X \times Y = Z_1$, a contradiction.

2.30

Suppose that

$$(X_0 \cap A) \subsetneq (X_1 \cap A) \subsetneq \cdots \subsetneq (X_n \cap A)$$

is a chain of closed sets X_i which defines a maximal chain in the subspace topology of A. Then consider the chain

$$\overline{X_0 \cap A} \subsetneq \overline{X_1 \cap A} \subsetneq \cdots \subsetneq \overline{X_n \cap A}$$

and note that these are irreducible by exercise 2.20b. Furthermore the containments remain proper by restricting to A and applying exercise 2.20a, getting back the chain we started with.

2.33

Write the matix in terms of its components

$$M = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

the condition of the rank being at most one is equivalent to all the columns being scalar multiples of eachother. Alternatively, it means that all the basic minors have determinant one, or that

$$ae - bd = 0 \qquad bf - ce = 0 \qquad af - cd$$

which defines an affine variety. The dimension of the variety is 4 since we have these degrees of freedom:

(a) a and d.

(b) The scalar multiples of $\begin{bmatrix} a \\ b \end{bmatrix}$ that defines the other two columns.

2.34

(a) Suppose that we have a chain of irreducible closed sets

$$Y_0 \subset Y_1 \subset \cdots \subset Y_n$$

Firstly choose an open set U in the cover intersecting Y_0. We claim that $Y_i \cap U \neq Y_{i+1} \cap U$ for every i so that

$$Y_0 \cap U \subset Y_1 \cap U \subset \cdots \subset Y_n \cap U$$

is a chain. For each i we can choose some open set U' so that $Y_i \cap U' \neq Y_{i+1} \cap U'$ since they form a cover. Therefore $Y_i^c \cap U' \cap Y_{i+1}$ is not empty (the c denotes complement). Furthermore it is open in the subspace topology, therefore it must intersect U in Y_{i+1} by irreducibility of Y_{i+1}. This proves the claim.

Now I need to prove the irreducibility of the terms in the chain. Suppose that U_1 and U_2 intersect $Y_i \cap U$. Then by irreducibility of Y_i, U_1, U_2, U all intersect eachother at some open set. Therefore U_1 and U_2 intersect inside of $Y_i \cap U$ as desired.

In the other direction we can apply 2.30.

(b) Note that in the proof of (a) the only condition on U was that it did intersect Y_0. Note that the length in an irreducible affine variety is the same no matter how the chain is constructed. Therefore choose any Y_0 to start the chain that intersects U.

2.35

This is by combining (a) and (b) of remark 2.31. Note that the proof of 2.31a still works when we restrict to chains that start with some Y_0 containing a.

2.36

(a) Consider some open cover U_i of X. The cover condition means that $\bigcap U_i^c = \emptyset$. Consider the chain

$$U_1^c \supseteq (U_1^c \cap U_2^c) \supseteq (U_1^c \cap U_2^c \cap U_3^c) \supseteq \cdots$$

and ignore the U_i which do not refine this chain at each step. By the Noetherian condition this chain must terminate (at which point the intersection is $\emptyset$ by construction) and we have obtained our finite subcover.

(b) Take any polynomial f in the coordinate ring of the variety and lift it to some representative in $\mathbb{A}^n$. It must achieve unbounded values by complex analysis, then by construction there is an unbounded continuous map $X \to \mathbb{C}$, a contradiction with compactness of X in the classical topology.

2.44

(a) Note that R is an integral domain by eisenstein on $x_1x_4 - x_2x_3$ considered as a polynomial in x_1. The constant term is in the prime ideal (x_2) but not the leading term. Therefore it is an integral domain since the polynomial is irreducible. Furthermore the dimension is 3 since the codimension of any principle prime ideal is 1 and $4 - 1 = 3$.

(b) By symmetry it suffices to show that x_1 is irreducible but not prime. Its not prime because if we quotient by (x_1) we get $K[x_2, x_3, x_4]/\langle x_2x_3 \rangle$ which is not an integral domain. It is irreducible because $\langle x_1x_4 - x_2x_3 \rangle$ is a homogeneous ideal. Therefore to satisfy the equation $fg = x_1 \pmod{x_1x_4 - x_2x_3}$ it suffices to check when f and g are homogeneous and their degrees add to 1. However then either f or g must have degree 0 and a unit.

(c) Note by (b) all of x_1, x_2, x_3, x_4 are irreducible and by the proof of (b) they are pairwise non-associate. Therefore these two expressions are decompositions into irreducible elements that do not agree.

(d) Firstly the ideal is prime since if we quotient it we get the integral domain $K[x_3, x_4]$. Note that

$$\dim_R(\langle x_1, x_2 \rangle) = \dim_{k[x_i]}(\langle x_1, x_2 \rangle) = 4 - 2 = 2$$

therefore the codimension is at most 1 since the dimension of R is 3 by (a). Furthermore if the ideal were principle then x_1 and x_2 would have a common factor that is not a unit. This is a contradiction with (b).

The Sheaf of Regular Functions

3.12

(a) Note that in a UFD we know that every prime ideal of codimension 1 is principle, and furthermore by Krull's principle ideal theorem every minimal prime ideal over an element is of codimension 1. In particular, principle prime ideals are minimal over their generator, hence codimension 1. Therefore the principal prime ideals are exactly those of codimension 1. We are reduced to proving that $\mathscr{O}_X(U) = A(X)$ iff $\mathcal{I}(Y)$ is not principle.

Firstly, if $\mathcal{I}(Y) = p_1$ then the function $1/p_1$ proves that $\mathscr{O}_X(U) \neq A(X)$. Otherwise suppose that $\mathcal{I}(Y) = (p_1, \ldots, p_n)$ for distinct irreducible elements. We choose two representative f/p_1^a on $D(p_1)$ and g/p_2^b on $D(p_2)$. By the condition that $fp_2^a = gp_1^b$ we know by the same reasoning as Example 3.11 that $a = 0$. Therefore $\varphi = f$ is a polynomial.

(b) Consider when $R = K[x_1, x_2, x_3, x_4]/\langle x_1x_4 - x_2x_3\rangle$ and set $Y = V(x_1, x_2)$. Then consider the regular function in Example 3.3 which is not a polynomial by that exposition.

3.20

Let $I = \mathcal{I}(X)$ and $R = \mathcal{I}(\mathbb{A}^n) = k[x_1, \ldots, x_n]$. By Defintion 3.19 we are essentially asked to prove that:

$$(R/I)_a \cong \frac{R_a}{IR_a}$$

It is a fact of commutative algebra that localization is exact and therefore commutes with quotients. This is proven in the commutative algebra note.

3.21

(a) This is a local ring with the maximal ideal $\{f : f(a) = 0\}$. It suffices to show that any element with $f(a) \neq 0$ is invertable. Since f is continuous we can restrict to an open set around a where f is nonzero. Then its inverse can be computed in this opens set as desired.

(b) This is not a local ring since any polynomial can be recovered globally by its local information at a point (the derivatives give the coefficients). Therefore in this case $\mathscr{F}_a$ is just the polynomial functions which is not a local ring.

3.22

(a) For each $a \in U$ find an open set U_α on which φ and ψ agree (this exists by the stalk condition). Now cover U with all the U_α. Note that φ and ψ restrict to the same functions on these open sets, therefore by the sheaf condition $\varphi = \psi$ since they have the same local information.

(b) If φ and ψ agree on one stalk then they agree on at least one open set. Then we can apply Remark 3.5 (Identity theorem for regular functions).

(c) Consider the sheaf of continuous functions as in Exercise 3.21. Now consider

$$f = x \qquad g = |x|$$

which agree on the open set $(0, \infty)$ but disagree at the stalk at $x = -1$ since they have different values there.

3.23

First note that that $D(h) \cap Y$ is nonempty iff $h \notin \mathcal{I}(Y)$ by unwrapping the definitions. Consider the morphism $\mathcal{A}(X)_{\mathcal{I}(Y)} \to \mathscr{O}_{X,Y}$ with $g/f \mapsto (D(f), g/f)$. This is valid by the claim and the fact that if $g/f = g'/f'$ in the localization then $h(gf' - g'f) = 0$ for some $h \notin \mathcal{I}(Y)$ so that in the open neighborhood $D(hf)$ we have $g/f = g'/f'$ as desired. The map is injective because if g/f maps to 0 then for some $h \in \mathcal{A}(X) \setminus \mathcal{I}(Y)$ we have $h(g \cdot 1 - 0 \cdot f) = 0$ but then $g = 0$ in the localization. The map is surjective since every function in a sufficiently small neighborhood is representable by a fraction g/f.

3.24

This is the statement that a direct limit is only effected by the later terms in the series. Alternatively we can define isomorphisms $\mathscr{F}(X)_a \leftrightarrow \mathscr{F}(U)_a$ with the maps in both directions given by:

$$(\phi, V) \mapsto (\phi, V \cap U)$$

These maps are well defined in the direct limit by the presheaf conditions. They are clearly inverses by unwrapping definitions so these rings are isomorphic.

Morphisms

4.12

We can write a general conic as:

$$ax^2 + by^2 + cxy + dx + ey + f$$

We first suppose via translation and affine transformation that the conic is 0 at the origin and moreover that the tangent line at 0 is x. This gives:

$$ax^2 + by^2 + cxy + ey$$

Consider what happens to the equation when we make the sub $x \mapsto x + \alpha y$:

$$ax^2 + (b + a\alpha^2 + c\alpha)y^2 + (2a\alpha + c)xy + ey$$

Now there are two cases.

Case $a = 0$: We get

$$(b + c\alpha)y^2 + cxy + ey$$

and note that c must be nonzero since the polynomial is irreducible. Therefore we can set $\alpha = -b/c$ to get $cxy + ey$. Now we take $y \mapsto y/c$ and $x \mapsto x - (1 + e)$ to get $xy - 1$ as desired.

Case $a \neq 0$: In this case we set $\alpha = -c/(2a)$ so that we obtain:

$$ax^2 + \left(b - \frac{c^2}{4a}\right) y^2 + ey$$

The middle term is very tricky. However, if we had applied the sub $y \mapsto \sqrt{\frac{c}{4ab}}y$ before starting this whole process we would have gotten a setup which leads to the middle term being 0. In this case e must be nonzero by the condition that the polynomial is irreducible. Now apply $x \mapsto x/\sqrt{a}$ and $y \mapsto y/e$ to obtain $x^2 - y$ as desired.

4.13

(a) The forward direction is true. Suppose that $a \neq b$. Then there is some $y \in Y$ such that $a(y) \neq b(y)$. Find some preimage $x \in X$ with $f(x) = y$ so that

$$f^*(a)(x) = a(f(x)) = a(y) \neq b(y) = b(f(y)) = f^*(b)(x)$$

On the other hand the backward direction is false. Consider the natural inclusion $k[x] \to \frac{k[x,y]}{(xy-1)} \cong k[x, x^{-1}]$. This map is injective, however the morphism of varieties it induces $(a, b) \mapsto (a)$ is not surjective since 0 has no preimage.

(b) The forward direction is false by Example 4.9. The backward direction is true. Suppose that f^* is surjective. Then $\mathcal{A}(X)$ is a quotient of $\mathcal{A}(Y)$, and its prime ideals correspond to the prime ideals of $\mathcal{A}(X)$ which contain the kernel of f^*. Therefore f is surjective since the maximal ideals are mapped to.

(c) This is true. A map $\mathbb{A} \to \mathbb{A}$ must be represented by a polynomial $f(x) = a_n x^n + \cdots + a_0$. However any polynomial of degree greater than 1 has multiple zeros and is therefore not injective.

(d) This is false. Consider the map $f(x, y) = (x + y^2, y)$ with inverse $f^{-1}(x, y) = (x - y^2, y)$.

4.19

First we examine the global sections to motivate our classification:

(a) Note that this is the distinguished open subset of $x - 1$. Therefore the ring of functions looks like:

$$\frac{k[x,y]}{xy-1} \cong k[x]_{x-1} \qquad x - 1 \mapsto x \qquad y \mapsto \frac{1}{x-1}$$

(b) The space of global sections is isomorphic to

$$\frac{k[x,y]}{x^2+y^2} \cong \frac{k[x,y]}{(x+iy)(x-iy)} \cong k[x] \times k[x]$$

using the Chinese Remainder Theorem.

(c) We note that this is isomorphic to $k[x]$ since we essentially are setting $x_2 = x_1^2$ and $x_3 = x_1^3$.

(d) Note that

$$\frac{k[x,y]}{xy} \cong k[x] \times k[x]$$

again by Chinese Remainder Theorem.

(e) Note that this polynomial is irreducible by considering it as a polynomial in y and applying Eisenstein on the prime ideal $x + 1$. Therefore we cannot simplify the ring $\frac{k[x,y]}{y^2-x^3-x^2}$.

(f) This is the equation for a rotated hyperbola. By applying a $\pi/4$ rotation this becomes $\frac{k[x,y]}{xy-1}$. This can be proven by doing the algebra or applying Exercise 4.12.

Therefore we may guess that (a) and (f) are isomorphic and that (b) and (d) are isomorphic. It suffices to show this since the others cannot be isomorphic since their global sections are not isomorphic. This is by the $\pi/4$ rotation for $(a) \leftrightarrow (f)$ and the map $f(x, y) = (\frac{x+y}{2}, \frac{x-y}{2i})$ for $(b) \leftrightarrow (d)$.

Varieties

5.7

(a) Consider some morphism $\mathbb{A}^1 \setminus \{0\} \to \mathbb{P}^1$. Recall that $\mathbb{P}_1$ is covered by two copies of $\mathbb{A}^1$ which we call U_1 and U_2. Consider the restriction of f to a map $f^{-1}(U_1) \to U_1 \cong \mathbb{A}_1$. By Proposition 4.7 this is a map of the form $\varphi_1 \in \mathscr{O}_{\mathbb{A}^1}(f^{-1}(U_1))$. Similarily we have a map $\varphi_2 \in \mathscr{O}_{\mathbb{A}^1}(f^{-1}(U_2))$. Now restrict to some open set $D(h)$ inside $f^{-1}(U_1) \cap f^{-1}(U_2)$ (which is nonempty since $\mathbb{A}^1$ is an irreducible space) where both of these maps are representable as fractions:

$$\varphi_1 = \frac{g_1}{h^{n_1}} \qquad \varphi_2 = \frac{g_2}{h^{n_2}}$$

Now by the combatibility of these maps on $U_1 \cap U_2$ we can draw the commuting diagram:

$$\begin{tikzcd} D(f) \arrow[r, "\varphi_1"] \arrow[d, "\varphi_2"] & U_1 \\ U_2 \arrow[ur, bend right, "x \mapsto \frac{1}{x}"'] & \end{tikzcd}$$

So that on $D(h)$ we have φ_2 agreeing with $\frac{h^{n_1}}{g_1}$. If we write φ_1 in lowest terms then it is impossible that both the numerator and denominator are in the prime ideal (x) by the UFD condition. Therefore either φ_1 or its reciprocal are defined at 0 and one of these allows the map to extend to 0.

(b) Consider the map $\mathbb{A}^2 \setminus \{0\} \to \mathbb{P}_1$ given by

$$(a, b) \mapsto \begin{cases} \frac{a}{b} \in U_1, & b \neq 0 \\ \frac{b}{a} \in U_2, & a \neq 0 \end{cases}$$

which is compatible on the overlap by checking and is a valid map on each open set by Proposition 4.7.

(c) In the case of a map $\mathbb{P}^1 \to \mathbb{A}^1$ we need two maps $f_1 : U_1 \to \mathbb{A}^1$ and $f_2 : U_2 \to \mathbb{A}^1$ which are compatible on the overlaps. Since $U_i \cong \mathbb{A}^1$ we can write f_1 and f_2 as elements of $k[x]$ by proposition 4.7. Then the compatibility on overlap condition gives the diagram:

$$\begin{tikzcd} & U_1 \cap U_2 \arrow[d, "f_1"] \\ U_2 \cap U_1 \arrow[ur, "x \mapsto \frac{1}{x}"] \arrow[r, "f_2"] & \mathbb{A}^1 \end{tikzcd}$$

Which tells us that $f_2(x) = f_1(x^{-1})$ on $U_1 \cap U_2 \cong \mathbb{A}^1 \setminus \{0\}$. Essentially they are equal in the localization $k[x]_x$. However the only way that these are both polynomials is if f_1 and f_2 are constant.

5.8

(a) The map $\mathbb{P}_1 \to \mathbb{P}_1$ restricts to four maps

$$\phi_{1a} : U_1 \cap f^{-1}(U_1) \to U_1 \qquad \phi_{1b} : U_1 \cap f^{-1}(U_2) \to U_2 \qquad \phi_{2a} : U_2 \cap f^{-1}(U_1) \to U_1 \qquad \phi_{2b} : U_2 \cap f^{-1}(U_2) \to U_2$$

which are compatible on the overlaps. The first thing we will do is consider the map:

$$\phi_{1a} : U_1 \cap f^{-1}(U_1) \to U_1$$

By the isomorphism property $f^{-1}(U_1)$ is all of $\mathbb{P}_1$ minus a single point. Therefore using $U_1 \cong \mathbb{A}_1$ the intersection $U_1 \cap f^{-1}(U_1)$ is a distinguished open set of the form $D(x + a)$ of $\mathbb{A}^1$. Then on this set we can write:

$$\phi_{1a} = \frac{f}{g} \qquad g = (x + a)^n$$

In the special case $f^{-1}(U_1) = U_1$ we just have $g = 1$ so I will merge the cases. Now suppose FTSOC that $\max(\deg f, \deg g) > 1$ and consider the expression:

$$\frac{f(x)}{g(x)} + b = \frac{f(x) + bg(x)}{g(x)}$$

We want to understand the roots of $f(x) + bg(x)$. Since we have written in lowest terms f does not vanish at any root of g (which is just a). Furthermore for all but at most one choice of b the expression will have degree $\max(\deg f, \deg g)$. Therefore it has multiple roots (counted with multiplicity) which correspond to points of $U_1 \cap f^{-1}(U_1)$ which ϕ_{1a} sends to $b \in U_1$. If there is a multiple root at some x_0 then we can write

$$b = -\frac{f(x_0)}{g(x_0)} = -\frac{f'(x_0)}{g'(x_0)} \implies f(x_0)g'(x_0) = f'(x_0)g(x_0)$$

This is a polynomial equation with finitely many roots. If we choose a b that is not $\frac{f(x_0)}{g(x_0)}$ for any root x_0 of this equation then roots of $f(x) = bg(x)$ will be distinct. This violates the injectivity of the map since the value b will be achieved multiple times. Therefore $\max(\deg f, \deg g) \leq 1$ and each of the four maps look like:

$$\frac{ax + b}{cx + d}$$

(b) It suffices to show that we can map any three distinct p, q, r to $0, 1, \infty$ (in this context ∞ is the unique point not in U_1). Then the desired map is by sending a_1, a_2, a_3 to $0, 1, \infty$ and applying the inverse of the map sending b_1, b_2, b_3 to $0, 1, \infty$. I also claim that the image of three points uniquely determine the map.

Consider the form of a Mobius transformation

$$\frac{ax + b}{cx + d}$$

By $r \mapsto \infty$ we have $d = -cr$ and by $p \mapsto 0$ we have $b = -ap$. Therefore write:

$$\frac{a}{c}\left(\frac{x - p}{x - r}\right)$$

We need $q \mapsto 1$ meaning $\frac{a}{c} = \frac{q-p}{q-r}$. The transformation is now determined up to scaling the numerator and denominator (which does not change the map).

As a fun remark, multiplying two Mobius transformations is analogous to multiplying the matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$$

which gives the new coefficients. Therefore the inverse of a Mobius transformation can be found by taking the matrix inverse.

5.9

Firstly note that every morphism $X \to Y$ gives a map $\mathscr{O}_Y(Y) \to \mathscr{O}_X(X)$ by definition. We need to determine whether a map $\mathscr{O}_Y(Y) \to \mathscr{O}_X(X)$ determines a unique morphism $X \to Y$.

(a) We claim that we can construct a unique map $X \to Y$. Consider the coordinate functions $y_i \in Y$. These will pull back to some regular functions $x_i \in X$. Then the map $p \mapsto (x_1(p), \ldots, x_n(p))$ is the desired map. We need to show that it is a valid morphism of ringed spaces. This is by Lemma 4.6, which states that it suffices to show the map is locally a morphism. We can do this by covering X with small open sets isomorphic to affine varieties, then apply Proposition 4.7 on the restricted maps.

(b) This direction is false. For a counterexample consider the maps $\mathbb{A}^1 \to \mathbb{P}^1$. Exercise 5.7c shows that $\mathscr{O}_Y(Y)$ is the constant functions. On the other hand, there are many more maps $\mathbb{A}^1 \to \mathbb{P}^1$ than just constant functions. A simple example is the embedding $\mathbb{A}^1 \subset \mathbb{P}^1$ using $U_1 \cong \mathbb{A}^1$.

5.11

Consider an open set $V \subseteq Y$ and a map $\varphi : V \to K$. I will show that the criterions in Construction 5.10b and the classical definition for φ to be a regular function in are equivalent. Then the isomorphism of ringed spaces is the trivial correspondance. Firstly we will denote:

$$R = \mathscr{O}_X(U) \qquad \overline{R} = \frac{R}{\mathcal{I}(Y)} \qquad x \in V$$

It is enough to show that the criterion are locally isomorphic.

5.10b $\implies$ Classical: Consider some principal open set $D(h)$ containing x for $h \in \overline{R}$. For any representative $\tilde{h}$ of h in R we can write by definition $D(\tilde{h}) \cap Y = D(h)$. Therefore if $\phi = f/h^n$ for $f \in \overline{R}$ in $D(h)$ then by lifting f to $\tilde{f} \in R$ we a representative $\tilde{f}/\tilde{h}^n$ on $D(\tilde{h})$ for φ in the sense of Construction 5.10.

Classical $\implies$ 5.10b: Consider some principal open set $D(h)$ containing x for $h \in R$ and denote $\overline{h}$ as the image of h under the quotient map. We can write by definition $D(h) \cap Y = D(\overline{h})$. Therefore if $\phi = f/h^n$ for $f \in R$ in $D(h)$ then by quotienting we obtain a local representative $\overline{f}/\overline{h}^n$ in the sense of the classical defintion.

5.21

First note that $\mathbb{P}^1 \times \mathbb{P}^1$ is covered by 4 open sets.

$$U_1 \times U_1 \qquad U_1 \times U_2 \qquad U_2 \times U_1 \qquad U_2 \times U_2$$

Consider the preimages of the diagonal ideal $\Delta_{\mathbb{P}^1}$. These will be sets of the form:

$$\{x \times x : x \in U_1\} \qquad \{x \times x^{-1} : x \in U_1 \cap U_2\} \qquad \{x^{-1} \times x : x \in U_1 \cap U_2\} \qquad \{x^{-1} \times x^{-1} : x \in U_2\}$$

We note that $U_i \cong \mathbb{A}^1$ and $U_1 \cap U_2 \cong \mathbb{A}^1 \setminus \{0\}$. Therefore in the first and fourth case we have something that looks like $\Delta_{\mathbb{A}_1}$ which is closed by Lemma 5.18a. In the second and third case we have something that looks like $\mathcal{V}(xy - 1) \subseteq \mathbb{A}^2$ which is closed. Therefore the diagonal of $\mathbb{P}^1$ is closed.

5.22

(a) We first note that the property of Proposition 5.20b characterized separated varieties. This is by considering the two projection maps $X \times X \to X$ whose agreement is the diagonal ideal.

Now consider two maps:

$$Z \underset{g}{\overset{f}{\rightrightarrows}} X \times Y$$

By the universal property of the product the maps are given by four maps:

$$f_1 : Z \to X \qquad f_2 : Z \to Y \qquad g_1 : Z \to X \qquad g_2 : Z \to Y$$

Now consider the agreement set of f and g. This will be the agreement set of f_1 and g_1 intersected with the agreement set of f_2 and g_2. Since X and Y are separated both these sets are closed and the intersection is closed as desired.

(b) Suppose that affine opens U_i cover X and V_i cover Y. Open subsets of irreducible spaces are irreducible so the U_i and V_i are irreducible. Now note that the $U_i \times V_j$ are irreducible by Exercise 2.24. Therefore $X \times Y$ is covered by irreducible open sets and is irreducible by Exercise 2.21b.

5.23

(a) Suppose that U and V are affine opens of X. Then note that $U \cap V$ can be identified with the intersection of $U \times V$ and Δ_X in $X \times X$. By Exercise 2.24 $U \times V$ is affine and now the intersection of an affine open with a closed set is affine.

(b) We identify $X \cap Y$ with $(X \times Y) \cap \Delta_X$ in $\mathbb{A}^n \times \mathbb{A}^n$. The decomposition of $X \times Y$ into irreducible component will be $X_i \times Y_j$ for the irreducible components of X_i of X and Y_j of Y. This is because they are distinct irreducible sets by Exercise 5.22b which cover $X \times Y$. Therefore $X \times Y$ is pure dimensional. Working in the coordinate ring $\mathcal{A}(X \times Y)$ note that Δ can be obtained as an intersection of n hypersurfaces.

$$\overline{x_1 - y_1}, \overline{x_2 - y_2}, \ldots, \overline{x_n - y_n}$$

where the bar denotes the image in $\mathcal{A}(X \times Y)$. Now by Krull Height theorem the codimension of Δ is at most n. By pure dimensionality of $X \times Y$ the dimension of the desired intersection is at least $\dim X + \dim Y - n$.

5.24

Take the irreducible decomposition $X = X_1 \cup \cdots \cup X_n$. Now note that by Remark 2.31a that $\dim X = \sup\{\dim X_i\}$. Furthermore $U = U \cap X_1 \cup \cdots \cup U \cap X_n$ is an irreducible decomposition of U so $\dim U = \sup\{\dim U \cap X_i\}$ and it suffices to prove that:

$$\dim U \cap X_i = \dim X_i$$

Therefore assume that X is irreducible. Since X is a prevariety it is covered by affine open sets U_i. Suppose that U_k is one such open set. Since U is dense it must intersect U_k at some U'_k. Then by Exercise 2.34b (and the fact that open sets of an irreducible space are irreducible) $\dim U'_k = \dim U_k$. Now note that U is covered by $U'_i = U \cap U_i$ so by Exercise 2.34a we may write

$$\dim X = \sup\{\dim U_i\} = \sup\{\dim U'_i\} = \dim U$$

as desired.

Note that this solution does not use the fact that X is a variety. I'm not sure what the intended solution is but my philosophy is to always try to prove the strongest statement available to me so I won't spend effort searching for this solution.

Projective Varieties I: Topology

6.8

$$\deg 1 = \deg(1 \cdot 1) = 2 \deg 1 \implies \deg 1 = 0$$

Note that this implies that the degree of any unit is 0 when $\mathbb{N}$ is the index set. If we instead use $\mathbb{Z}$ as the index set then this is no longer true.

6.14

Suppose that $a = (a_0 : \cdots : a_n)$ and suppose WLOG that $a_0 \neq 0$. Then we can construct the ideal:

$$\langle a_0 x_1 - a_1 x_0, a_0 x_2 - a_2 x_0, \ldots, a_0 x_n - a_n x_0 \rangle$$

Now suppose that $p = (p_1, \ldots, p_n)$ is a point satisfying this polynomial. Each term expresses that $a_0 p_i = p_0 a_i$ so that $p_i = \frac{p_0}{a_0} a_i$ and p and a have the same homogeneous coordinates up to scaling by the factor $\frac{p_0}{a_0}$ as desired.

6.29

First consider a line in $\mathbb{P}^3$. A line is given by the intersection of two hypersurfaces

$$L = \mathcal{V}(\langle a_1x_1 + b_1x_2 + c_1x_3 + d_1x_4, a_2x_1 + b_2x_2 + c_2x_3 + d_2x_4\rangle)$$

by Remark 6.28. Written in this way, we can see that lines in $\mathbb{P}^3$ are in correspondance with planes in $\mathbb{A}^4$ (if we give it the structure of a vector space) via projectivization.

Their intersection will therefore be a line in $\mathbb{A}^4$. A point in $\mathbb{P}^1$ corresponds to a line in $\mathbb{A}^4$. The statement reduces to the fact that in $\mathbb{A}^4$ there is a plane two arbitrary lines. This is true due to linear algebra.

6.30

(a) Suppose that we have integrality for homogeneous elements. Then for any two elements f and g the lowest degree term of fg will be the product of the lowest degree terms. Therefore it will be nonzero and $fg \neq 0$.

(b) Suppose that $fg = 0$. Then note

$$\mathcal{V}(f) \cup \mathcal{V}(g) = \mathcal{V}(fg) = \mathcal{V}(0) = X$$

so that $X = \mathcal{V}(f) \cup \mathcal{V}(g)$ is reducible. In the other direction suppose that $X = \mathcal{V}(I) \cup \mathcal{V}(J)$. Then $X = \mathcal{V}(IJ)$ so by the Nullstellensatz $IJ = 0$. Therefore for nonzero elements $f \in I$ and $g \in J$ we have $fg = 0$.

6.31

(a) Suppose that

$$\mathcal{I}_a(C(X)) \subseteq \mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n \subset \mathfrak{p}_{n+1} = \langle x_0, \ldots, x_n\rangle$$

is a chain of not neccesarily homogeneous ideals. We claim that we can construct a maximal chain of equal length of homogeneous ideals ending at $\langle x_0, \ldots, x_n\rangle$, this will then prove that the homogeneous dimension of the cone is one less then its affine dimension.

Firstly we note that $\mathfrak{p}_0$ is homogeneous due to the result that any minimal prime ideal in a graded ring (for instance $R/\mathcal{I}(X)$) is homogeneous. This is because $\mathfrak{p}_0^h$ is a prime ideal contained in $\mathfrak{p}_0$ but still containing $\mathcal{I}(X)$ (since it contains its homogeneous generators), therefore $\mathfrak{p}_0 = \mathfrak{p}_0^h$ by minimality.

Now we will induct, suppose that we have modified the chain so that $\mathfrak{p}_0 \subset \cdots \subset \mathfrak{p}_k$ is homogeneous. Choose some homogeneous element x which is not contained in $\mathfrak{p}_k$.

Claim 6.0.1. We can modify the chain so that $\mathfrak{p}_{k+1}$ contains x.

Proof. We induct on r, the maximal index so that $\mathfrak{p}_r$ does not contain x. Apply the Krull principal ideal theorem on minimal primes over x in the ring $(R/\mathfrak{p}_{r-1})_{\mathfrak{p}_{r+1}}$. This yields a prime ideal which replaces $\mathfrak{p}_r$ in the chain. It will be nonzero since it contains x and not $\mathfrak{p}_{r+1}$ since it has codimension 1. ☐

Now let $\mathfrak{p}'_{k+1}$ be the ideal generated by the homogeneous elements of $\mathfrak{p}_{k+1}$. Note that $\mathfrak{p}'_{k+1}$ is contained in $\mathfrak{p}_{k+1}$, but not equal to $\mathfrak{p}_k$ since it contains x. Therefore we can replace $\mathfrak{p}_{k+1}$ with $\mathfrak{p}'_{k+1}$ in the chain (these must be equal if the chain is maximal). Then by the induction we can modify the chain so that every ideal is homogeneous.

(b) Note that $C(X \cap Y) = C(X) \cap C(Y)$ so by (a) we can write:

$$\dim(X \cap Y) = \dim(C(X) \cap C(Y)) - 1$$

And by applying 5.23b it follows that the dimension of an irreducible component is atleast:

$$\dim(X \cap Y) = \dim(C(X) \cap C(Y)) - 1 \geq \dim C(X) + \dim C(Y) - (n+1) - 1 = \dim X + \dim Y - n$$

and by the proof of that theorem an irreducible component must exist in the case that $\dim X + \dim Y - n \geq 0$. Therefore the intersection is nonempty.

6.36

I am not going to attempt to make tikz render this thing. Check out desmos, it makes a lot of sense why the points are where they are when you see the graph.

We can homogenize the polynomial into:

$$x_1^3 - x_1 x_2^2 + x_0^3$$

When $x_0 = 0$ we have $x_1^2 = x_2^2$ or $(x_1 - x_2)(x_1 + x_2)$. In projective space this corresponds to the points $[1 : 1 : 0]$ and $[1 : -1 : 0]$.

Projective Varieties II: Ringed Spaces

7.3

(a) We have shown that the new $\mathbb{P}^1$ is covered by two open sets U_1 and U_2 both isomorphic to $\mathbb{A}^1$. These are $\{[x : 1]\}$ and $\{[1 : x]\}$ To show the equivalence we need to consider the intersection. Those will be the points with homogeneous coordinates $[x, y]$ with x and y nonzero. In each U_i this corresponds to all points except 0. Furthermore the maps can be constructed

$$[x : 1] \sim [1 : 1/x] \qquad [1 : x] \sim [1/x : 1]$$

as desired.

(b) Consider a function $\varphi : U \to k$, we want to show that the criterion for φ to be a regular function on ϕ is the same for the old and new defintion. It suffices to check locally. Therefore we can restrict to an affine open subset of U_1 (by symmetry). Then the result follows from Proposition 7.2 and Exercise 5.11.

7.6

Consider the isomorphism in Example 7.5:

$$X = \mathcal{V}(x_0 x_2 - x_1^2) \qquad f : X \to \mathbb{P}^1, (x_0 : x_1 : x_2) \mapsto \begin{cases} (x_1 : x_2), & \text{if } (x_0 : x_1 : x_2) \neq (1 : 0 : 0) \\ (x_0 : x_1), & \text{if } (x_0 : x_1 : x_2) \neq (0 : 0 : 1) \end{cases}$$

We claim that the coordinate ring are not isomorphic. This is because $\frac{k[x_0, x_1, x_2]}{(x_0 x_2 - x_1^2)}$ is not a UFD (since $x_0 x_2$ and x_1^2 are two ways to write the same element, x_0, x_1, x_2 are all irreducible). On the other hand $S(\mathbb{P}^1)$ is a UFD by the Guass lemma.

7.7

I will assume the result of Exercise 7.8 because it makes stuff easier. The proof of that exercise does not rely on this one.

(a) I first claim that the hypersurfaces on $\mathbb{P}^n$ are the sets of the form $\mathcal{V}(f)$ for some $f \in S(\mathbb{P}^n)_d$. By Exercise 6.31a hypersurfaces in $\mathbb{P}^n$ correspond to hypersurfaces in $\mathbb{A}^{n+1}$ under the cone operation. These cones will all be of the form $\mathcal{V}(f)$, so that the projectivizations $\mathbb{P}(\mathcal{V}(f))$ will be generated by a single element.

Now suppose for the sake of contradiction that A hypersurface in $\mathbb{P}^m$ is given by the zero locus of some homogeneous $f \in S(\mathbb{P}^m)$. By Exercise 7.8 f will pull back to some $\tilde{f}$ along the map. Therefore the inverse image of the hypersurface is $\mathcal{V}(\tilde{f})$, a hypersurface by the claim.

(b) Take the $m+1$ hypersurfaces $x_0, \dots, x_m$ in $\mathbb{P}^m$ and note that their intersection is $\emptyset$. By (a) these will pull back to $m+1$ hypersurfaces $V_0, \dots, V_m$ in $\mathbb{P}^n$. Note that by what was proved Exercise 6.31b the intersection of a hypersurface with any variety V has dimension at least $V - 1$. Therefore the intersection of $V_0, \dots, V_m$ will have dimension at least $n - m - 1$ and therefore be nonempty when $n > m$. This is a contradiction since we cannot have points mapping to nothing.

(c) There is no nonconstant map $\mathbb{P}^{n+m} \to \mathbb{P}^n$ by (b). On the other hand the universal property of the product gives a projection map $\mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^n$.

7.8

First we will prove an anologue of Proposition 4.7. Suppose that $f : U \to Y$ is a morphism where U is a an open subset of a projective variety and $Y \subseteq \mathbb{P}^m$ is a projective variety. Now consider the restrictions

$$f_i : f^{-1}(U_i) \to U_i$$

It suffices to consider these since that $f^{-1}(U_i)$ must cover U since the U_i cover Y. Note that on this open set the pullback of x_j/x_i is a regular function which we will call $\varphi_{ij} \in \mathscr{O}_{\mathbb{P}^n}(f^{-1}(U_i))$. Locally each φ_{ij} is representable by a fraction of homogeneous polynomials a_{ij}/b_{ij} of the same degree. Now we have that on $f^{-1}(U_0)$:

$$f_0 : p \mapsto (1 : \varphi_{01}(p) : \cdots : \varphi_{0m}(p))$$

therefore locally we can write

$$f_0 : p \mapsto \left(1 : \frac{a_{01}}{b_{01}} : \cdots : \frac{a_{0m}}{b_{0m}}\right)$$

and by scaling all the coordinates by $b_{01} \cdots b_{0m}$ we obtain that f is locally given coordinate wise by polynomials of the same degree.

Now consider a map $f : \mathbb{P}^n \to \mathbb{P}^m$. Take a cover of $\mathbb{P}^n$ by distinguished open sets $U(h_i)$ on which f is given coordinate wise by polynomials of the same degree. Now note that $\bigcap \mathcal{V}(h_i) = \emptyset$ meaning $\langle h_i \rangle$ contains 1 and we can find some subset $h_1, \dots, h_r$ so that $a_1 h_1 + \cdots + a_r h_r = 1$. Therefore $\langle h_1, \dots, h_r \rangle = R$ and we obtain a finite cover of $\mathbb{P}^n$ by distinguished open sets $U(h_1), \dots, U(h_r)$. Now we may write

$$f|_{D(h)} = (f_0 : \cdots : f_m) \qquad f|_{D(h')} = (f'_0 : \cdots : f'_m)$$

By multiplying f_i through by a power of h we may assume that the f_i vanish on $\mathcal{V}(h)$ and likewise for the f'_i. In particular this means that

$$f_i f'_j = f'_i f_j$$

so that in the fraction field we can write

$$\frac{f_1}{f'_1} = \cdots = \frac{f_n}{f'_n}$$

which will equal some a/b when written in lowest terms. By the UFD condition each f_i/a can be written as a polynomial (not a fraction), and the function

$$f|_{D(h) \cup D(h')} = (f_0/a : \cdots : f_m/a)$$

is well defined. Therefore since we have a finite cover of $\mathbb{P}^n$ we can find a global representative of f.

7.14

First we claim that the closed sets in $X \times Y$ are generated by polynomials which are homogeneous in the x_i and homogeneous in the y_i. Suppose that $f(x_i, y_i)$ is one such polynomial. We define f_k to be the polynomial given by replacing x_i with z_{ik} and y_i with z_{ki}. Then f_k is a polynomial in the z_i and defines a closed set in the Segre embedding. However we can write

$$f_k \sim x_k^{\deg f} f$$

so that f being 0 on a point is equivalent to all the f_k being 0 on a point. Therefore the zero locus of f is a closed set in the Segre embedding. In the other direction if f is a polynomial in the z_{ij} just take $z_{ij} \mapsto x_i y_j$.

Now we will show that the complement of this set is closed. Firstly, the fact that the diagonal is closed shows that the set of distinct points $a \times b$ is an open subset. Now consider the conditon that the line through a, b does not meet the cubic at a third point. By Bezout's theorem this is equivalent to a or b intersecting the line with multiplicity. This will happen when either a or b is a multiple point or the line through a, b is the tangent line through a or b. The multiplicity occurs when a or b is in the intersection of f and all its derivatives f_X, f_Y, f_Z. In the other case the tangent to a point on a cubic curve f on $p \in \mathbb{P}^2$ is the line:

$$f_x(p)x + f_y(p)y + f_z(p)z$$

Therefore the condition that a is a multiple point or that b is on the tangent line to a is that the polynomial

$$f_{x_0}y_0 + f_{x_1}y_1 + f_{x_2}y_2$$

is 0 on $a \times b$. This cuts out a closed set by the claim. Then by symmetry, the complement of the desired set is:

$$\Delta_{\mathbb{P}^2} \cap (\mathcal{V}(f_{x_0}y_0 + f_{x_1}y_1 + f_{x_2}y_2) \cup \mathcal{V}(f_{y_0}x_0 + f_{y_1}x_1 + f_{y_2}x_2))$$

Since all three terms are closed the set is closed as desired.

7.15

In terms of a direct argument note that $\mathbb{P}^1 \times \mathbb{P}^1 \cong \mathbb{P}^2/(z_{00}z_{11} - z_{10}z_{01})$. Then by all the same argumentation at Exercise 2.40 $\langle z_{00}, z_{10} \rangle$ is a (homogeneous) prime ideal of codimension 1 that is not principle.

For a more structural result, we claim that hypersurfaces in a projective variety X are exactly those of the form $\mathcal{V}(f)$ iff the homogeneous coordinate ring is a UFD.

Note that it suffices to prove the result for irreducible hypersurfaces. Recall by Exercise 6.31 that a homogeneous prime ideal $\mathfrak{p}$ being of codimension 1 (a hypersurface) is equivalent to being codimension 1 in $C(X)$. Since X and $C(X)$ have the same coordinate rings the result follows from the affine case (this is Proposition 2.37).

7.28

(a) The variables in the Veronese embedding of degree d are indexed by the monomials of degree d, for instance $z_{x_1x_2x_3}$ is one if $d = 3$. The image condition is:

$$z_{f_1}z_{f_2} - z_{g_1}z_{g_2} \text{ for all monomials } f_1, f_2, g_1, g_2 \text{ of degree } d \text{ with } f_1f_2 = g_1g_2$$

It is clear that the image into the Veronese coordinates satisfies these condtions. Now suppose that all the z_m satisfy the condition. Note that the image is covered by the open sets where $x_i \neq 0$. It suffices to consider inside of each of these open sets. Therefore suppose WLOG that $z_{x_0^d} = 1$ and write

$$x_i = z_{x_i x_0^{d-1}}$$

which gives a set of coordinates that must map to our set of coorinates. This is because

$$x_{i_1} \cdots x_{i_d} \mapsto z_{x_{i_1}x_0^{d-1}} \cdots z_{x_{i_d}x_0^{d-1}} = z_{x_{i_1},\dots,x_{i_d}} z_{x_i^d}^{d-1} = z_{x_{i_1},\dots,x_{i_d}}$$

by the relations we have imposed.

(b) Suppose that X is a projective variety. It is described as $\mathcal{V}(\langle f_1, \dots, f_n \rangle)$ for some homogeneous polynomials. Note that by Remark 6.23 we may assume WLOG that $f_1, \dots, f_n$ have the same degree $2d$. Then the Veronese embedding of degree d sends the f_i to quadratic polynomials. Furthermore the relations found in (a) are quadratic. Therefore X can be described as the 0 locus of only quadratic polynomials in a very large space.

7.30

First note that if X_1 and X_2 are complete then $X_1 \times X_2$ is complete since the projection can be broken up

$$X_1 \times X_2 \times Y \overset{\pi_1}{\to} X_2 \times Y \overset{\pi_2}{\to} Y$$

and the composite map has all the desired properties.

(a) Using the coefficents of a quadric we can express an isomorphism between the space of quadric curves and $\mathbb{P}^5$.

$$ax_0^2 + bx_1^2 + cx_2^2 + dx_0x_1 + ex_0x_2 + fx_1x_2 \mapsto (a : b : c : d : e : f)$$

We can also express an isomorphism between the space of linear curves and $\mathbb{P}^2$:

$$ax_0 + bx_1 + cx_2 \mapsto (a : b : c)$$

Now note that a quadric curve being reducible is equivalent as being able to write it as a product of linear polynomials. Therefore define the map

$$\mathbb{P}^2 \times \mathbb{P}^2 \to \mathbb{P}^5 \qquad (a : b : c) \times (a' : b' : c') \mapsto (aa' : bb' : cc' : ab' + a'b : ac' + a'c : bc' + b'c)$$

whose image is isomorphic to the space of reducible quadrics. This map is a well defined morphism of varieties since the polynomials are bihomogenous (we can make a similar argument as Proposition 7.2). Note that $\mathbb{P}^2 \times \mathbb{P}^2$ is complete by the claim. Therefore its image is closed by Corollary 7.23 and avoiding its image defines an open subset of $\mathbb{P}^5$.

(b) The condition that a conic is 0 at some $p = (p_0 : p_1 : p_2)$ defines a linear equation on the coefficients by plugging in:

$$ap_0^2 + bp_1^2 + cp_2^2 + d(p_0p_1) + e(p_0p_2) + f(p_1p_2) = 0$$

(c) By (b) the condition of intersecting each point defines a hyperplane in the space of quadrics. Therefore the condition of intersecting 5 points gives 5 linear equations constraining the space of solutions. This will give a unique conic when the constraints are not linearily dependent. To prove the the constraints are linearily independent when the points are distinct, it suffices to construct a quadric passing through $P_1, \dots, P_4$ but not P_5. This is the product of the lines through P_1, P_2 and P_3, P_4, which does not pass through P_5 since no 3 points are collinear.

Therefore, there is a unique quadric passing through these points. Furthermore it is not reducible since then it would be a product of two lines, and by pigeonhole 3 of the points must lie on the same line.

7.31

(a) In this context f is the morphism given by forgetting the 3rd coordinate:

$$(y_0 : y_1 : y_2 : y_3) \mapsto (y_0 : y_1 : 0 : y_3) \in L$$

Therefore the coordinate map is given by the inclusion: Furthermore the image of the Veronese embedding is all the points of the form:

$$(x_0^3 : x_0^2 x_1 : 0 : x_1^3)$$

We note that x_0^3 and x_1^3 can be chosen independently by picking x_0 and x_1 to be one of their cube roots. Note that the equation $y_1^3 = y_0^2 y_3$ is satisfied, we claim this condition is equivalent to being in the image. If y_0 or y_3 is 0 then $y_1 = 0$ by the equation and $(\sqrt[3]{y_0} : \sqrt[3]{y_3})$ hits this point. Otherwise suppose WLOG that $y_3 = 1$ and $y_1 \neq 0$ and note

$$x_0 = y_0/y_1 \qquad x_1 = 1$$

hits these points by doing some algebraic manipulation with the equation $y_1^3 = y_0^2 y_3$:

(b) This is not an isomorphism onto its image. The image $\mathcal{V}(y_1^3 - y_0^2 y_3, y_2)$ has a singular point at $(0 : 0 : 0 : 1)$. This means that the local ring at that point is not a regular local ring, however the local ring of its preimage is. This is a contradiction with the map being an isomorphism.

Grassmannians

As a quick remark, I dislike the convention used in the section. In order to mesh more seamlessly with general linear algebra intuition I feel like an element of $G(k, n)$ should be thought of as a $n \times k$ matrix with trivial kernel modulo column operations. Then the ideas like the row span become the column span, which is more natural in my opinion (this is just the image of the matrix).

8.22

(a) By lemma 8.14 it suffices to show that this is the condition for the map $v \mapsto v \wedge \omega$ to have rank 2. Firstly suppose that $\omega = \sum a_{ij} v_{ij}$. We index a basis of $\wedge^3 V$ as $v_{234}, v_{134}, v_{124}, v_{123}$ and write the matrix of the map:

$$\begin{bmatrix} 0 & a_{34} & -a_{24} & a_{23} \\ a_{34} & 0 & -a_{14} & a_{13} \\ a_{24} & -a_{14} & 0 & a_{12} \\ a_{23} & -a_{13} & a_{12} & 0 \end{bmatrix}$$

By permuting $v_1, \ldots, v_4$ we may assume WLOG that a_{12} is nonzero. Therefore the rank two condition is equivalent to the first two columns being linear combinations of the last two. By matching the 3rd and 4th coordinates and then checking the 1st and 2nd. We get the same equation:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} \tag{1}$$

(b) We think of L as an elementary tensor $\ell = v_1 \wedge v_2$. Another line $\rho = w_1 \wedge w_2$ will intersect this line iff the subspaces they represent overlap. By lemma 8.10 this is equivlant to $\ell \wedge \rho = 0$. If we write a basis v_1, v_2, v_3, v_4 for V and write $\rho = \sum r_{ij} a_{ij} v_{ij}$ then by computing this means that $a_{34} = 0$. This equation remains linear in any system of coordinates and moreover its coefficients satisfy Eq. (1).

Each line defines a hyperplane in $\mathbb{P}^5$ by (b). In general these hyperplanes will intersect to a line by counting dimensions. We now must find the number of intersections with this line and

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23}$$

which is a quadric surface. By paramaterizing the line we see that in general it is intersected twice. Therefore there are in general two unique lines which intersect 4 given lines.

8.23

(a) We work in the Plucker embedding. Consider a line $\ell = v_1 \wedge v_2$ and a point v. The condition of being incident means that $\ell \wedge v = 0$. Therefore the incidence correspondance is the kernel of the map $\ell \times v \to \ell \wedge v$. If we choose a basis then this map is a bihomogenous polynomial in the coordinates of ℓ and v, therefore defining a closed subvariety when intersected with $x_{12}x_{34} - x_{13}x_{24} + x_{14}x_{23}$.

(b) In this solution I will implicitly use the results that projective varieties are complete and separated. Futhermore these properties are preserved under closed subsets and products.

Denote that incidence correspondance as $I \subseteq G(2,n) \times \mathbb{P}^{n-1}$. Define $I_Z = I \cap (G(2,n) \times Z)$ to be the lines intersecting Z and their points, this is closed when Z is closed. Now consider the two projections:

$$I_X \times I_Y \overset{\pi_1}{\underset{\pi_2}{\rightrightarrows}} G(2,k)$$

By separatedness of I_Z the agreement of these two morphisms is a closed set $R \subseteq I_X \times I_Y$. An element of this agreement is $(L \times p_1) \times (L \times p_2)$ where L intersects X and Y and p_1, p_2 lie on L. Tracing definitions we see that R is a subset of $(G(2,n) \times \mathbb{P}^{n-1}) \times (G(2,n) \times \mathbb{P}^{n-1})$. The image of R under one of the projections onto $\mathbb{P}^{n-1}$ is a closed set since $G(2,n) \times \mathbb{P}^{n-1} \times (G(2,n)$ is complete. By construction this set is exactly the join and we are done.

Birational Maps and Blowing Up

9.9

Let f be this polynomial. Firsly, by making a projective change of coordinates we may assume that $(1 : 0 : \cdots : 0)$ lies on the curve and is not a multiple point. We can write $f = f_1 x_0 + f_2$ where f_1, f_2 are polynomials in $x_1 \ldots x_n$. On the distinguished open set $D(f_1)$ the projection map $\mathbb{P}^n \to \mathbb{P}^{n-1}$ by $(x_0, \ldots, x_n) \mapsto (x_1, \ldots, x_n)$ is birational with inverse $(x_1, \ldots, x_n) \to (-(f_2/f_1)(x_1, \ldots, x_n)), x_1, \ldots, x_n)$. This is because $f = 0 \implies x_0 = -f_2/f_1$.

9.18

Recall the equality:

$$\widetilde{X} \subseteq \mathcal{V}(y_i f_j(x) - y_j f_i(x)) \cap (X \times \mathbb{P}^{r-1}) \qquad \Longrightarrow \qquad \widetilde{\mathbb{A}^3} \subseteq \mathcal{V}(y_1 x_2 - y_2 x_1) \cap (\mathbb{A}^3 \times \mathbb{P}^1)$$

We claim that this inluclusion is an isomorphism. Consider the open subset where $y_1 = 1$. Then we have $x_2 = y_2 x_1$. Therefore consider

$$(x_0, x_1, y_2 x_1) \mapsto (x_0, x_1, y_2 x_1) \times (x_1, y_2) \sim (x_0, x_1, x_2) \times (1, y_1)$$

which implies that the map is an isomorphism. Now we must show that its exceptional set is isomorphic to $\mathbb{A}^1 \times \mathbb{P}^1$. Since the exceptional set is where $x_1 = x_2 = 0$ the condition $y_1 x_2 - y_2 x_1 = 0$ becomes trivial and we just get

$$\mathcal{V}(x_1, x_2) \cap (\mathbb{A}^3 \times \mathbb{P}^1)$$

which is just $\mathbb{A}^1 \times \mathbb{P}^1$ as desired.

9.19

The open sets $Y_1 \setminus Y_2$ and $Y_2 \setminus Y_1$ in Y_1 and Y_2 respectively are disjoint. Therefore they become two disjoint open sets in the blowup since it acts as an isomorphism on $X \setminus (Y_1 \cap Y_2)$. We need to show that the closures are disjoint in the blowup, write

$$\mathcal{I}(Y_1) = \langle f_1, \ldots, f_n \rangle \qquad \mathcal{I}(Y_2) = \langle g_1, \ldots, g_r \rangle$$

and consider the blowup at $\langle f_i, g_i \rangle$. Note that $y_{f_i} = 0$ and $y_{g_i} = 0$ is satisfied on Y_1 and Y_2 respectively. Thereofore on $\widetilde{Y}_1$ and $\widetilde{Y}_2$ (their closures) we have $y_{f_i} = 0$ and $y_{g_i} = 0$ respectively so that $y_{f_i} = y_{g_i} = 0$ on $\widetilde{Y}_1 \cap \widetilde{Y}_2$. This is a contradiction since $(0 : \cdots : 0)$ does not exist in $\mathbb{P}^{n+r-1}$.

9.20

Schemes

12.21

One direction is clear. Therefore suppose that $\operatorname{Spec} R = \mathcal{V}(I) \cup \mathcal{V}(J)$. Then note that $\operatorname{Spec} R = \mathcal{V}(IJ)$ so IJ is contained in the nilradical of R. Furthermore since $\mathcal{V}(I + J) = \mathcal{V}(I) \cap \mathcal{V}(J) = \emptyset$ it follows that $1 \in \sqrt{I + J} \implies I + J = R$ so I and J are comaximal. Therefore take $1 = x + y$ so that $x \in I$ and $y \in J$. We know that $x^n y^n = 0$ for some n. Note using binomial coefficients expand $1 = (x + y)^{2n}$ to find $a \in (x^n)$ and $b \in (y^n)$ so that

$$a + b = 1 \qquad ab = 0$$

since $ab \in (x^n y^n) = (0)$. These are projections as desired. Alternatively one could use CRT on (a) and (b).